

Department Overview Department Name: Department of Computer Engineering Faculty:
 Faculty of Information & Communication Technology (FICT) Accreditation: The program is
 fully accredited by the Pakistan Engineering Council (PEC). Core Focus: Blends hardware
 engineering and computer science concepts. It covers software programming, hardware design,
 industrial automation, embedded systems, robotics, multimedia, and artificial intelligence. Vision
 and Mission Vision: To produce professional Computer Engineers who can use knowledge,
 technical expertise, and entrepreneurial abilities to compete with upcoming challenges in
 computing fields while adhering to highest standards of morality and ethics. Mission: To prepare
 graduates to accept and fulfill responsibilities across a broad spectrum of activities, including
 analysis, synthesis and design of the modern computing systems, serve government and industry
 in various roles and contribute in academia and research. Bachelor of Science (BS) Program
 Admission Requirements Academic Background: F.Sc (Pre-Engineering) / ICS with
 Mathematics & Physics, or an equivalent qualification with at least 60% marks. Alternatively, a
 Diploma of Associate Engineering (DAE) in a relevant field with at least 60% marks. Entry Test:
 Candidates must clear an entrance test conducted by NTS. Degree Requirements Total Credit
 Hours: 134 Total Courses: 43 Total Labs: 21 or 26 Minimum Academic Standing: CGPA $\geq$ 2.0 Curriculum Plan (Program Schema by Semester) 1st Semester CS 118/118L Computer
 Fundamentals & Lab Prerequisite: None Credit Hours: 2+1 PHY 205/205L Applied Physics &
 Lab Prerequisite: None Credit Hours: 2+1 MATHP 118 Calculus & Analytical Geometry
 Prerequisite: None Credit Hours: 3 HUM 101 Islamic Studies/Ethics Prerequisite: None Credit
 Hours: 2 ENG 161 Reading and Writing Skills (Eng.-1) Prerequisite: None Credit Hours: 3
 ENGG 202L Workshop Practice Lab Prerequisite: None Credit Hours: 1 ENVIRON102
 Occupational Health and Safety Prerequisite: None Credit Hours: 1 2nd Semester CS 112/112L
 Computer Programming & Lab Prerequisite: None Credit Hours: 3+1 MATHP 111 Linear
 Algebra Prerequisite: None Credit Hours: 2 EE 221/221L Electronic Circuits and Devices & Lab
 Prerequisite: None Credit Hours: 3+1 ENG 361 Communication Skills Prerequisite: None Credit
 Hours: 2 EE 203/203L Linear Circuit Analysis & Lab Prerequisite: None Credit Hours: 3+1 3rd
 Semester HUM 102 Pakistan Studies and Global Perspective Prerequisite: None Credit Hours: 2
 TE 305/305L Computer Comm. and Networks & Lab Prerequisite: None Credit Hours: 3+1 CS
 212/212L Object-Oriented Programming & Lab Prerequisite: Computer Programming + Lab
 Credit Hours: 3+1 MATHA 213 Complex Variables and Transforms Prerequisite: None Credit
 Hours: 3 EE 110/110L Digital Logic Design & Lab Prerequisite: None Credit Hours: 3+1
 MATHA 234 Discrete Structures Prerequisite: None Credit Hours: 3 4th Semester EE 261/261L
 Signals & Systems & Lab Prerequisite: Complex Variables and Transforms Credit Hours: 3+1
 CE 211/211L Computer Architecture and Organization & Lab Prerequisite: Digital Logic Design
 + Lab Credit Hours: 3+1 MATHA 214 Differential Equations Prerequisite: Calculus &
 Analytical Geometry Credit Hours: 3 CS 214/214L Data Structures & Algorithms & Lab
 Prerequisite: Object-Oriented Programming + Lab Credit Hours: 3+1 TBA MS-Elective I
 Prerequisite: None Credit Hours: 3 5th Semester CE 326/326L Microprocessors and Interfacing
 & Lab Prerequisite: Computer Architecture and Org. + Lab Credit Hours: 3+1 CS 341/341L
 Operating Systems & Lab Prerequisite: Object-Oriented Programming + Lab Credit Hours: 3+1
 TBA Social Science-Elective I Prerequisite: None Credit Hours: 2 HUM 292 Technical Writing
 Prerequisite: None Credit Hours: 3 TE 412/412L Digital Signal Processing & Lab Prerequisite:
 Signals & Systems + Lab Credit Hours: 3+1 6th Semester TBA CEDE - I & CEDE - II
 Prerequisite: TBA Credit Hours: Var (2+1 or 3) MATHA 217 Probability Methods in
 Engineering Prerequisite: None Credit Hours: 3 CS 331/331L Database Management Systems &

Lab Prerequisite: None Credit Hours: 3+1 EE 302/302L Digital System Design & Lab
Prerequisite: Digital Logic Design + Lab Credit Hours: 3+1 7th Semester TBA MDEE - I &
CEDE-III Prerequisite: TBA / None Credit Hours: Var TBA / TBA Numerical Analysis & Lab
Prerequisite: TBA Credit Hours: 2+1 CE 451L Senior Design Project-I Prerequisite: None Credit
Hours: 3 CE 303/303L Software Engineering & Lab Prerequisite: None Credit Hours: 2+1 8th
Semester TBA CEDE - IV Prerequisite: TBA Credit Hours: Var TBA Management Science
Elective-II Prerequisite: None Credit Hours: 3 TBA MDEE - II Prerequisite: None Credit Hours:
3 CE 452L Senior Design Project-II Prerequisite: Senior Design Project-I Credit Hours: 3 TBA
Social Science Elective-II Prerequisite: None Credit Hours: 3 Master of Science (MS) Program
Admission Requirements Academic Background: 16 years of education or equivalent (e.g., B.E
or BS 4-year degree) in a relevant field from an HEC-recognized university. Applicants must
have at least 60% marks under the annual grading system or a minimum CGPA of 2.5 out of 4.0.
Entry Test: Must clear the admission test conducted by the University or its authorized testing
agency with a minimum cumulative score of 50%. Degree Requirements Total Credit Hours: 30
Total Courses: 10 Minimum Academic Standing: CGPA $\geq$ 3.0